Functional Annotation Report: *prpC* (PP_2335 / UniProt Q88KF5) — 2-Methylcitrate Synthase of *Pseudomonas putida* KT2440

Summary

The gene **prpC** (ordered locus **PP_2335**, UniProt accession **Q88KF5**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes **2-methylcitrate synthase (2-MCS; EC 2.3.3.5)**, a soluble, cytoplasmic enzyme belonging to the citrate synthase superfamily. Its primary function is to catalyze the **committed condensation step of the 2-methylcitrate cycle**: the Claisen-type condensation of **propionyl-CoA with oxaloacetate (and water)** to yield **(2S,3S)-2-methylcitrate plus free coenzyme A**. Although the enzyme retains the ancestral ability to condense acetyl-CoA with oxaloacetate to form citrate, kinetic characterization of orthologs shows that **propionyl-CoA is the strongly preferred substrate**, distinguishing PrpC from the housekeeping citrate synthase of the tricarboxylic acid (TCA) cycle.

This report confirms the gene/protein identity with high confidence. The gene symbol *prpC*, the ordered locus PP_2335, the InterPro domain architecture (2-MeCitrate/Citrate_synth_II, IPR011278, plus the canonical citrate-synthase fold domains), the KEGG orthology assignment (K01659), and the genomic context of the gene (embedded within a *prp* operon, PP_2333–PP_2338) all converge on a single, unambiguous conclusion. A global sequence alignment demonstrates that the P. putida protein is **53.6% identical** to the biochemically characterized and crystallized *Salmonella* Typhimurium 2-methylcitrate synthase (StPrpC), with the two catalytic residues (His261 and Asp312) mapping exactly onto their StPrpC counterparts. In contrast, PrpC shares only **35.3% identity** with its own organism's housekeeping citrate synthase (GltA/PP_4194), placing it firmly in the specialized 2-MCS clade rather than the general citrate-synthase clade.

Physiologically, PrpC initiates the pathway by which *P. putida* oxidizes propionate and propionyl-CoA — a metabolite that is cytotoxic when it accumulates — into pyruvate and succinate that can enter central metabolism. In *P. putida*, this methylcitrate cycle is the **only known route for propionate oxidation**. The enzyme functions in the cytoplasm, requires no metal cofactor for the condensation chemistry itself, and — unlike some Gram-negative citrate synthases — is **not allosterically regulated by NADH**, its regulation instead being imposed at the transcriptional level through the *prp* operon. The following report details the evidence for each of these conclusions and situates PrpC within the broader biochemistry of the 2-methylcitrate cycle.

Gene/Protein Identity Verification

Before presenting functional findings, the mandatory identity check is addressed directly. All independent lines of evidence agree:

Verification criterion	Expected (from UniProt)	Finding	Match
Gene symbol	prpC	Consistent with *prp* (propionate) catabolic locus	□
Ordered locus	PP_2335	Sits in *prp* operon PP_2333–PP_2338	□
Organism	*P. putida* KT2440	Confirmed	□
Protein family	Citrate synthase family	IPR011278 (2-MeCitrate/Citrate_synth_II) + citrate-synthase fold domains	□
Catalytic identity	Citrate synthase (RecName)	Dual citrate synthase / 2-methylcitrate synthase; 2-MCS is the physiological role	□
KEGG orthology	—	K01659 (2-methylcitrate synthase, EC 2.3.3.5)	□

The gene symbol *prpC* is **not** ambiguous in this case: it corresponds to the well-established *prp* (propionate catabolism) nomenclature used across *Salmonella*, *Corynebacterium*, *Pseudomonas*, and other bacteria. Literature retrieved describes this exact gene family and its function, and the organism-specific features of the *P. putida* operon are consistent with related pseudomonads. **Research proceeded with high confidence on the correct target.**

Key Findings

Finding 1 — PrpC is 2-methylcitrate synthase catalyzing propionyl-CoA + oxaloacetate → 2-methylcitrate + CoA

The InterPro/UniProt domain assignment places PP_2335 in the **methylcitrate-synthase subfamily** of the citrate synthase superfamily: it carries IPR011278 ("2-MeCitrate/Citrate_synth_II") together with the citrate-synthase fold domains (IPR016142, IPR016143, IPR002020, and the citrate-synthase active-site signature IPR019810). This domain architecture is the sequence-level signature of enzymes that perform Claisen condensation of an acyl-CoA thioester with oxaloacetate.

The functional assignment rests on **direct biochemical proof in orthologs**. In *Salmonella* Typhimurium, the PrpC enzyme was purified to homogeneity and shown to have 2-methylcitrate synthase activity in vitro, forming 2-methylcitrate from propionyl-CoA and oxaloacetate (PMID: 10482501). In *Corynebacterium glutamicum*, characterization of two *prpDBC* gene clusters revealed that their gene products were active as citrate synthases and 2-methylcitrate synthases (PMID: 11976302). The gene name *prpC* and its operon position (PP_2335, adjacent to *prpB*, *prpD*, and the *acnD/prpF* dehydratase-isomerase genes) match the canonical *prp* propionate-catabolic locus, so the P. putida gene is confidently assigned the same catalytic function.

> *"the PrpC enzyme was purified to homogeneity and shown to have 2-methylcitrate synthase activity in vitro"* — PMID: 10482501
>
> *"revealed that their gene products were active as citrate synthases and 2-methylcitrate synthases"* — PMID: 11976302

Finding 2 — Substrate specificity: propionyl-CoA is strongly preferred over acetyl-CoA and butyryl-CoA

A defining property of PrpC that distinguishes it from housekeeping citrate synthase is its substrate preference. Kinetic characterization of purified *Salmonella* PrpC established that although the enzyme *can* use acetyl-CoA as a substrate to synthesize citrate, kinetic analysis demonstrated that **propionyl-CoA is the preferred substrate** (PMID: 10482501). A later study of the crystallized *Salmonella* enzyme (StPrpC) confirmed that it **utilizes propionyl-CoA more efficiently than acetyl-CoA or butyryl-CoA** (PMID: 20970504). Specificity is conferred not by the catalytic residues themselves — which are identical to those of citrate synthase — but by **residues adjacent to the active site** (e.g., Tyr197 and Leu324 in StPrpC) that shape the acyl-CoA binding pocket to accommodate the extra methyl group of propionyl-CoA.

> *"Although PrpC could use acetyl-CoA as a substrate to synthesize citrate, kinetic analysis demonstrated that propionyl-CoA is the preferred substrate"* — PMID: 10482501
>
> *"StPrpC was found to utilize propionyl-CoA more efficiently than acetyl-CoA or butyryl-CoA"* — PMID: 20970504

Finding 3 — PrpC operates in the 2-methylcitrate cycle that detoxifies propionyl-CoA and converts propionate to pyruvate

PrpC catalyzes the **second, committed step** of the 2-methylcitrate cycle. In *Salmonella*, ^{13}C-labeling reconstruction assigned each Prp enzyme's role in an ordered pathway: **PrpE** (propionyl-CoA synthetase) activates propionate to propionyl-CoA; **PrpC** (2-methylcitrate synthase) condenses it with oxaloacetate to form 2-methylcitrate; **PrpD** (2-methylcitrate dehydratase) and aconitase/**AcnA** (2-methylisocitrate hydratase) carry out the dehydration/rehydration; and **PrpB** (2-methylisocitrate lyase) cleaves 2-methylisocitrate to pyruvate and succinate, oxidizing propionate's α-carbon to pyruvate (PMID: 10482501; PMID: 11294638).

In *Pseudomonas*, the same cycle is the principal route for propionate oxidation. Disrupting the downstream PrpB (2-methylisocitrate lyase) step blocks completion of the cycle (PMID: 37995793), and in *P. aeruginosa* the 2-methylcitrate and glyoxylate cycles are enzymatically linked and partially redundant (PMID: 40015638). The broader physiological purpose is **detoxification**: fungal 2-methylcitrate synthase (mcsA) is described as being responsible for detoxifying propionyl-CoA, a toxic metabolite (PMID: 31141475).

> *"The reactions are catalyzed by propionyl coenzyme A (propionyl-CoA) synthetase (PrpE), 2-methylcitrate synthase (PrpC), 2-methylcitrate dehydratase (probably PrpD), 2-methylisocitrate hydratase (probably PrpD), and 2-methylisocitrate lyase (PrpB)"* — PMID: 10482501
>
> *"responsible for detoxifying propionyl-CoA, a toxic metabolite"* — PMID: 31141475

Finding 4 — Structure and mechanism: conserved citrate-synthase fold, cytoplasmic, domain-closure catalysis, not NADH-regulated

The 2.4 Å X-ray crystal structure of *Salmonella* StPrpC established that the **polypeptide fold and catalytic residues are conserved with citrate synthases**, implying shared catalytic mechanism. The enzyme is dimeric (forming higher oligomers at high concentration) and adopts a nearly closed conformation, consistent with the induced-fit **domain-closure mechanism** characteristic of citrate synthases, in which a large domain movement sequesters substrates from solvent during catalysis. Critically, **the key residues involved in allosteric NADH binding are not conserved in StPrpC** — meaning PrpC is not subject to the NADH feedback inhibition seen in hexameric Gram-negative citrate synthases (PMID: 20970504).

Comparative crystallography of *Aspergillus fumigatus* mcsA versus human citrate synthase reinforced this picture: the active sites are near-identical, with substrate specificity determined by residues near the active site and by enzyme conformational changes (PMID: 31141475). Consistent with the citrate synthase family, PrpC is a **soluble cytoplasmic protein** with no signal peptide or transmembrane features; it carries out its function in the bacterial cytoplasm where propionyl-CoA and oxaloacetate are generated.

> *"The polypeptide fold and the catalytic residues of StPrpC are conserved in citrate synthases (CSs) suggesting similarities in their functional mechanisms"* — PMID: 20970504
>
> *"the key residues involved in NADH binding are not conserved in StPrpC"* — PMID: 20970504

Finding 5 — Bioinformatic confirmation: PP_2335 = 2-methylcitrate synthase with conserved catalytic His261/Asp312

UniProt entry Q88KF5 (375 amino acids, ~42 kDa) annotates **two catalytic activities**: acetyl-CoA + oxaloacetate → citrate (citrate synthase reaction) and propanoyl-CoA + oxaloacetate + H_2O → (2S,3S)-2-methylcitrate + CoA (2-methylcitrate synthase reaction), with pathway assignments to the TCA cycle and "propanoate degradation." Two catalytic active-site residues are annotated by the PIRSR rule PIRSR001369: **His261** (within a conserved G-F-G-H motif) and **Asp312** — the acid/base histidine and catalytic aspartate that are hallmarks of the citrate synthase superfamily active site. KEGG independently assigns PP_2335 to orthology group **K01659** (2-methylcitrate synthase, EC 2.3.3.5). These sequence-based assignments (evidence codes ECO:0000256 PIRNR/PIRSR) independently corroborate the biochemical identity established in orthologs.

Finding 6 — The *P. putida* KT2440 *prp* operon uses an AcnD/PrpF-type architecture with a GntR-family regulator

The genomic neighborhood of PP_2335 defines a complete methylcitrate-cycle gene cluster:

Locus	Gene	Product	KO	EC
PP_2333	—	GntR-family transcriptional regulator	—	—
PP_2334	prpB	2-methylisocitrate lyase	K03417	4.1.3.30
PP_2335	**prpC**	**2-methylcitrate synthase**	**K01659**	**2.3.3.5**
PP_2336	acnD	2-methylcitrate dehydratase (2-methyl-*trans*-aconitate forming)	K20455	4.2.1.117
PP_2337	prpF	2-methylaconitate isomerase	K09788	5.3.3.–
PP_2338	prpD	2-methylcitrate dehydratase	K01720	4.2.1.79

This clustering places PrpC as the **committed condensation enzyme** that feeds the downstream isomerase/lyase steps. Notably, *P. putida* uses the **AcnD** + **PrpF** (2-methyl-*trans*-aconitate isomerase) branch of the cycle rather than the single PrpD-aconitase route of *Salmonella*, and its operon regulator is **GntR-family** rather than the σ^{54}/RpoN-dependent PrpR activator of *Salmonella*. In *Salmonella*, the *prp* regulator PrpR is a member of the sigma-54 family of transcriptional activators (PMID: 10648513); the contrast highlights organism-specific regulation of an otherwise conserved catabolic module.

> *"members of the sigma-54 family of transcriptional activators"* — PMID: 10648513

Finding 7 — Physiological rationale and evolution: detoxification-plus-salvage; 2-MCS and citrate synthase share a common ancestor

The methylcitrate cycle converts cytotoxic propionyl-CoA to pyruvate; when the cycle is impaired, toxic intermediates (propionyl-CoA, 2-methylcitrate, 2-methylisocitrate) accumulate and inhibit numerous enzymes, including dehydrogenases (PMID: 37764443). This dual role — detoxification of a poisonous metabolite plus salvage of its carbon skeleton into central metabolism — is the raison d'être of PrpC.

Evolutionarily, **2-methylcitrate synthase and citrate synthase evolved from a common ancestral protein**, as did isocitrate lyase and 2-methylisocitrate lyase (PMID: 37764443; PMID: 31141475). PrpC is thus a gene-duplication-derived specialist that acquired preferential propionyl-CoA activity while retaining the ancestral fold and catalytic chemistry. In *P. putida*, the cycle is the **only known and described pathway for propionate oxidation** (PMID: 37995793); blocking a downstream step (PrpB) is non-lethal but redirects propionyl-CoA into polyhydroxyalkanoate (PHA) polymer synthesis, a finding exploited biotechnologically to fine-tune PHA monomer composition.

> *"the methylcitrate cycle converts cytotoxic propionyl-coenzyme A (CoA) to pyruvate"* — PMID: 37764443
>
> *"Fungal citrate synthase and 2-methylcitrate synthase as well as isocitrate lyase and 2-methylisocitrate lyase, each evolved from a common ancestral protein"* — PMID: 37764443
>
> *"the only known and described pathway for propionate oxidation in this organism"* — PMID: 37995793

Finding 8 — Sequence homology: 53.6% identity to characterized *Salmonella* 2-MCS with exact catalytic-residue conservation

A global Needleman–Wunsch alignment (BLOSUM62) of *P. putida* KT2440 PrpC (Q88KF5, 375 aa) against the biochemically characterized and crystallized *Salmonella* Typhimurium 2-methylcitrate synthase StPrpC (Q56063, 389 aa; EC 2.3.3.5) yields **201/375 = 53.6% identity**. Crucially, the two annotated catalytic residues of the P. putida enzyme map exactly onto the catalytic residues of StPrpC: **Ppu His261 ↔ Stm His274** and **Ppu Asp312 ↔ Stm Asp325**. Because StPrpC is precisely the enzyme shown to prefer propionyl-CoA over acetyl-CoA/butyryl-CoA and whose crystal structure defined the 2-MCS fold (PMID: 20970504; PMID: 10482501), this high identity and exact active-site conservation transfer the experimentally validated function to PP_2335 with strong confidence.

> *"The polypeptide fold and the catalytic residues of StPrpC are conserved in citrate synthases (CSs) suggesting similarities in their functional mechanisms"* — PMID: 20970504

Finding 9 — PrpC is a specialized paralog distinct from housekeeping citrate synthase GltA (PP_4194)

P. putida KT2440 encodes two citrate-synthase-family proteins: **PrpC** (Q88KF5/PP_2335, 375 aa) and the **housekeeping citrate synthase GltA** (Q88FA4/PP_4194, 429 aa). Pairwise Needleman–Wunsch (BLOSUM62) alignment shows that PrpC shares only **35.3% identity (132/374)** with its own organism's GltA, but **53.6% identity** with the biochemically validated *Salmonella* 2-MCS StPrpC. This clean separation — greater similarity to the validated 2-MCS ortholog than to the co-encoded housekeeping enzyme — demonstrates that **PrpC groups phylogenetically with the 2-MCS clade rather than the housekeeping citrate-synthase clade**, and that the two enzymes have specialized, non-overlapping metabolic roles: GltA drives the TCA cycle with acetyl-CoA, while PrpC drives the methylcitrate cycle with propionyl-CoA.

Mechanistic Model / Interpretation

PrpC (PP_2335) is the **gatekeeper enzyme of propionate detoxification** in *Pseudomonas putida* KT2440. The coherent narrative that emerges from the nine findings is as follows.

The reaction. In the cytoplasm, PrpC catalyzes a Claisen condensation identical in chemistry to citrate synthase but tuned for a larger acyl donor:

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propionyl-CoA + oxaloacetate + H2O  --PrpC-->  (2S,3S)-2-methylcitrate + CoA-SH
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The reaction proceeds through the conserved citrate-synthase mechanism: the catalytic **His261** acts as a general acid/base to enolize the acyl-CoA thioester, the enolate attacks the carbonyl of oxaloacetate, and **Asp312** participates in proton shuttling, followed by hydrolysis of the citryl-CoA-analog intermediate to release CoA and the free acid product. A large domain-closure motion sequesters the reaction from bulk solvent, as seen in the StPrpC crystal structure.

Why propionyl-CoA and not acetyl-CoA. The catalytic residues are identical to those of citrate synthase, so specificity is imposed by the *shape of the acyl-binding pocket* — active-site-adjacent residues that create room for the extra methyl group of propionyl-CoA. This is an elegant example of substrate discrimination achieved without altering the catalytic machinery, and it explains why PrpC is a physiologically dedicated 2-MCS even though it retains residual citrate-synthase activity.

Where it sits in metabolism.

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Propionate
   |  PrpE (propionyl-CoA synthetase)
   v
Propionyl-CoA  ────────────────► [toxic if it accumulates]
   |  PrpC (2-methylcitrate synthase)   ◄── THIS ENZYME (PP_2335)
   |  + oxaloacetate
   v
(2S,3S)-2-methylcitrate
   |  PrpD / AcnD (2-methylcitrate dehydratase)
   v
2-methyl-cis-aconitate ─ PrpF (2-methylaconitate isomerase) ─► 2-methyl-trans-aconitate
   |  aconitase / AcnA (2-methylisocitrate hydratase)
   v
2-methylisocitrate
   |  PrpB (2-methylisocitrate lyase)
   v
PYRUVATE  +  SUCCINATE  ──► central carbon metabolism (TCA cycle)
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Two roles at once. By consuming propionyl-CoA, PrpC simultaneously (i) **detoxifies** a metabolite that otherwise inhibits dehydrogenases and other enzymes, and (ii) **salvages** the carbon into pyruvate and succinate for growth. Propionyl-CoA arises in *P. putida* from diverse sources — degradation of odd-chain fatty acids, catabolism of certain amino acids (Ile, Val, Met, Thr), and breakdown of monoterpenes such as β-myrcene (in related pseudomonads). The methylcitrate cycle is the sole exit route for this pool in *P. putida*.

Regulatory logic. PrpC is controlled at the level of transcription of its operon rather than by allosteric feedback: it lacks the NADH-binding residues that regulate some Gram-negative citrate synthases. In *P. putida* the operon carries a **GntR-family regulator (PP_2333)**, in contrast to the σ^{54}-dependent PrpR of enteric bacteria — but the underlying logic is the same: the enzymes are induced when propionate (or its methylcitrate-derived signal) is present. In *Salmonella*, 2-methylcitrate itself (the product of PrpC) or a derivative serves as the inducing coregulator of the operon (PMID: 9851993).

Distinct from the TCA-cycle citrate synthase. The presence of a separate housekeeping citrate synthase (GltA/PP_4194) with only 35% identity confirms a clean division of labor. GltA handles acetyl-CoA condensation for the TCA cycle; PrpC handles propionyl-CoA condensation for the methylcitrate cycle. This paralog separation is the genomic signature of a dedicated propionate-detoxification module.

Property	PrpC (PP_2335)	GltA (PP_4194)
Enzyme	2-methylcitrate synthase	Citrate synthase (housekeeping)
Preferred acyl-CoA	Propionyl-CoA	Acetyl-CoA
Product	(2S,3S)-2-methylcitrate	Citrate
Pathway	2-methylcitrate cycle	TCA cycle
Length	375 aa	429 aa
% identity to Salmonella StPrpC	53.6%	(housekeeping clade)
Identity to each other	35.3%	35.3%

Evidence Base

PMID	Title (abbreviated)	How it supports the findings
10482501	*Salmonella typhimurium LT2 catabolizes propionate via the 2-methylcitric acid cycle*	Direct biochemical proof: purified PrpC has 2-MCS activity; propionyl-CoA is the preferred substrate; defines the ordered pathway enzymes. Foundation for F001, F002, F003.
20970504	*Crystal structure of Salmonella typhimurium 2-methylcitrate synthase*	Structure/mechanism: conserved fold & catalytic residues, domain closure, propionyl-CoA > acetyl-CoA > butyryl-CoA, NADH-binding residues NOT conserved. Foundation for F002, F004, F008.
11976302	*Two prpDBC clusters in Corynebacterium glutamicum*	Confirms PrpC orthologs act as citrate/2-methylcitrate synthases in a second organism. Supports F001.
11294638	*In vitro conversion of propionate to pyruvate (PrpD, aconitase)*	Reconstitutes the full 2-methylcitrate cycle in vitro; clarifies downstream enzyme roles that PrpC feeds. Supports F003.
31141475	*Aspergillus fumigatus 2-MCS vs human citrate synthase*	Establishes detoxification role of propionyl-CoA; near-identical active sites; specificity from active-site-adjacent residues. Supports F003, F004, F007.
10648513	*prpR, ntrA, ihf required for prpBCDE expression*	Salmonella regulator is σ^{54}-family — contrasts with the GntR-family regulator in the P. putida operon. Supports F006.
9851993	*Regulation of the prpBCDE operon*	Shows 2-methylcitrate (from PrpC) or a derivative is the inducing signal for the operon. Contextual support for regulation.
37995793	*Fed-batch cultivation with glucose+propionate (PHA composition)*	Methylcitrate cycle is the only propionate-oxidation route in Pseudomonas; blocking PrpB redirects flux to PHA. Supports F003, F007.
40015638	*Methylcitrate & glyoxylate cycles linked in P. aeruginosa*	Enzymatic redundancy/crosstalk between the cycles in a close relative. Supports F003.
37764443	*Methylcitrate cycle crosstalk in pathogenic fungi (review)*	Detoxification of cytotoxic propionyl-CoA; common evolutionary ancestor of CS and 2-MCS. Supports F007.

Additional retrieved literature (PMID: 36377867, PMID: 16879647, PMID: 19798672, PMID: 39492536, PMID: 40020280) provides broader context on propionate/organic-acid metabolism and propionyl-CoA toxicity in *Pseudomonas*, *Mycobacterium*, and related organisms, reinforcing the physiological importance of the methylcitrate cycle that PrpC initiates. In particular, the β-myrcene catabolism study in *Pseudomonas* sp. M1 shows the methylcitrate cycle channeling propionyl-CoA from a degradation pathway into central metabolism (PMID: 19798672).

Limitations and Knowledge Gaps

1. **No direct biochemical characterization of the *P. putida* KT2440 enzyme.** The function of PP_2335 is assigned by strong homology (53.6% identity, exact catalytic-residue conservation) to biochemically validated orthologs (*Salmonella* StPrpC) and by genomic context, not by purification and kinetic assay of the P. putida protein itself. No published K_m, $k$cat, or crystal structure exists specifically for Q88KF5.
2. **Substrate specificity ratios are transferred, not measured.** The quantitative preference for propionyl-CoA over acetyl-CoA/butyryl-CoA is documented for orthologs; the exact kinetic constants for the P. putida enzyme are unknown, though the conserved active-site-adjacent residues make the qualitative preference highly likely.
3. **Regulatory details in P. putida are incompletely defined.** The operon carries a GntR-family regulator (PP_2333) rather than a σ^{54} activator, but the precise inducing signal, promoter architecture, and regulon in KT2440 have not been experimentally dissected here.
4. **Oligomeric state in P. putida is inferred.** StPrpC is dimeric (with higher-order oligomers at high concentration); the native quaternary structure of the P. putida enzyme has not been directly determined.
5. **Localization is inferred bioinformatically.** The cytoplasmic assignment is based on the absence of signal/transmembrane features and the known behavior of the citrate synthase family; it has not been experimentally confirmed for this protein.

Proposed Follow-up Experiments / Actions

1. **Heterologous expression and kinetic characterization.** Clone PP_2335, express with an affinity tag, purify, and measure steady-state kinetics (K_m, *k*cat, catalytic efficiency) for propionyl-CoA, acetyl-CoA, and butyryl-CoA with oxaloacetate to confirm the predicted specificity ratio in the P. putida enzyme.
2. **Structural determination.** Solve the crystal (or cryo-EM) structure of the P. putida enzyme, ideally with bound substrate/product analogs, to verify the domain-closure mechanism and map the acyl-CoA pocket residues (analogous to StPrpC Tyr197/Leu324) that dictate propionyl-CoA preference. AlphaFold modeling followed by structural superposition on StPrpC (PDB) would be a rapid first step.
3. **Gene deletion and growth phenotyping.** Construct a clean ΔprpC (ΔPP_2335) mutant and test growth on propionate (and odd-chain fatty acids) as sole carbon source; a growth defect specific to propionate — rescued by complementation — would confirm the physiological role, and metabolomics would reveal propionyl-CoA/2-methylcitrate accumulation.
4. **Regulatory dissection.** Characterize the GntR-family regulator (PP_2333): identify its operator, the inducing metabolite (2-methylcitrate vs propionate), and the promoter driving PP_2333–PP_2338, using reporter fusions and EMSA.
5. **Confirm distinct roles of PrpC vs GltA.** Test whether GltA (PP_4194) can complement a ΔprpC mutant on propionate and vice versa, to establish functional non-redundancy of the two citrate-synthase-family paralogs.
6. **Biotechnological exploitation.** Given that flux through the methylcitrate cycle competes with propionyl-CoA incorporation into PHA polymers, modulate PrpC expression to tune odd-chain-length monomer content of medium-chain-length PHAs in engineered *P. putida* strains.

Conclusion

prpC (PP_2335, UniProt Q88KF5) of *Pseudomonas putida* KT2440 encodes **2-methylcitrate synthase (EC 2.3.3.5)**, a soluble cytoplasmic enzyme of the citrate-synthase superfamily. It catalyzes the committed condensation of **propionyl-CoA with oxaloacetate (and water) to form (2S,3S)-2-methylcitrate plus CoA**, using conserved catalytic residues His261 and Asp312 and strongly preferring propionyl-CoA over acetyl-CoA. Through this reaction it initiates the **2-**

methylcitrate cycle, the pathway that oxidizes propionate/propionyl-CoA to pyruvate and succinate while detoxifying growth-inhibitory propionyl-CoA. In *P. putida* the gene lies in an AcnD/PrpF-variant *prp* operon (PP_2333–PP_2338) under GntR-family transcriptional control rather than NADH allostery, and it is a specialized paralog clearly distinct from the housekeeping TCA-cycle citrate synthase GltA (PP_4194).
